

Name: _____

Exam Style Questions

Laws of Indices



Equipment needed: Pen, Calculator

Guidance

1. Read each question carefully before you begin answering it.
2. Check your answers seem right.
3. Always show your workings

Video Tutorial

www.corbettmaths.com/contents

Videos 17, 174



Answers and Video Solutions



1.  $2^2 \times 2^3 = 2^{\boxed{5}}$

Write the missing power in the box.

(1)

2. Simplify $5^4 \times 5^3$



Circle the correct answer.

5^7

25^7

5^{12}

5^7

25^{12}

(1)

3. Simplify $n^2 \times n^4$



n^6

(1)

4. $p^6 \times p^2 = p^x$



Find the value of x

$p^6 \times p^2 = p^8$

8

(1)

5. Jonah was asked to simplify $y^8 \times y^3$



His answer is y^{24}

Explain Jonah's mistake.

Jonah has multiplied the powers, he should have added them. y^{11}

(1)

6.



(a) Simplify

$$w^4 \times w^4$$

$$w^8$$

(1)

(b) Simplify

$$a^6 \div a^3$$

$$a^3$$

(1)

7.



(a) Simplify

$$m^5 \times m^3$$

$$m^8$$

(1)

(b) Simplify

$$m^8 \div m^2$$

$$m^6$$

(1)

(c) Simplify

$$(m^3)^2$$

$$m^6$$

(1)

8. Simplify $\frac{c^{12}}{c^4}$



$$c^8$$

(1)

9. Simplify

$$\frac{m^9 \times m}{m^5}$$



$$\frac{m^{10}}{m^5}$$

$$m^5$$

(2)

10. Match the expressions.



$$c^{10} \div c^5$$

$$c^{16}$$

$$c^7 \times c^3$$

$$c^5$$

$$(c^8)^2$$

$$c^2$$

$$\frac{c^6}{c^4}$$

$$c^{10}$$

(4)

11. Simplify the following.



$$\frac{s^3 \times s^4}{s^2}$$

$$\frac{s^7}{s^2}$$

$$s^5$$

(2)

12.



(a) Simplify $w^3 \times w^{-5}$

$$w^{-2}$$

(1)

(b) Simplify $a^4 \div a^{-2}$

$$a^6$$

(1)

13. Simplify $c^7 \times (c^6)^4$



$$c^7 \times c^{24}$$

$$c^{31}$$

(2)

14. Work out the value of $\frac{2^9 \times 2^{-2}}{2^3}$



$$\frac{2^7}{2^3} = 2^4$$

$$2 \times 2 \times 2 \times 2$$

$$16$$

(2)

15. Simplify $(5^{-3} \times 5^7)^3$



Give your answer as a power of 5.

$$(5^4)^3 = 5^{12}$$

$$5^{12}$$

(2)

16. Simplify $2a^3c^3 \times 3a^2c$



$$6a^5c^4$$

$$6a^5c^4$$

(2)

17. Simplify



$$\frac{10m^5n^4}{2m^2n}$$

$$5m^3n^3$$

(2)

18. Simplify $(7x^3)^2$



$$49x^6$$

Circle the correct answer.

$$49x^5$$

$$14x^6$$

$$14x^5$$

$$49x^6$$

(1)

19. Simplify $(2m^4)^3$



$$8m^{12}$$

$$8m^{12}$$

(2)

20. Simplify $(2xy^3)^4$



$$16x^4y^{12}$$

(2)

21. Work out the value of $3^{16} \div (3^2)^7$



$$3^{16} \div 3^{14} = 3^2$$

9

(2)

22. Simplify



$$\frac{a^{\frac{1}{5}} \times a^{\frac{2}{3}}}{a^{\frac{3}{5}}}$$

$$\frac{1}{5} + \frac{2}{3} = \frac{13}{15}$$

$$\frac{a^{\frac{13}{15}}}{a^{\frac{3}{5}}}$$

$$\frac{13}{15} - \frac{3}{5} = \frac{4}{15}$$

$$a^{\frac{4}{15}}$$

$a^{\frac{4}{15}}$

(3)

23. Simplify fully



$$\sqrt{\frac{50\pi a^5}{2\pi a^3}}$$

$$\sqrt{25a^2} = 5a$$

$$5a$$

(3)

24. Work out



$$\sqrt[3]{\frac{2^3 \times 3^6}{3^2}}$$

$$\sqrt[3]{2^3 \times 3^6}$$

$$= 2 \times 3^2$$

$$= 2 \times 9 = 18$$

$$18$$

(2)

25. Simplify $25 \times 5^3 \times 5^6$



Give your answer as a power of 5.

$$5^2 \times 5^3 \times 5^6 = 5^{11}$$

$$5^{11}$$

(2)

26. Write 27×9^x as a power of 3 in terms of x



$$3^3 \times (3^2)^x$$

$$3^3 \times 3^{2x}$$

$$3^{2x+3}$$

$$3^{2x+3}$$

(3)

or

$$3^{3+2x}$$